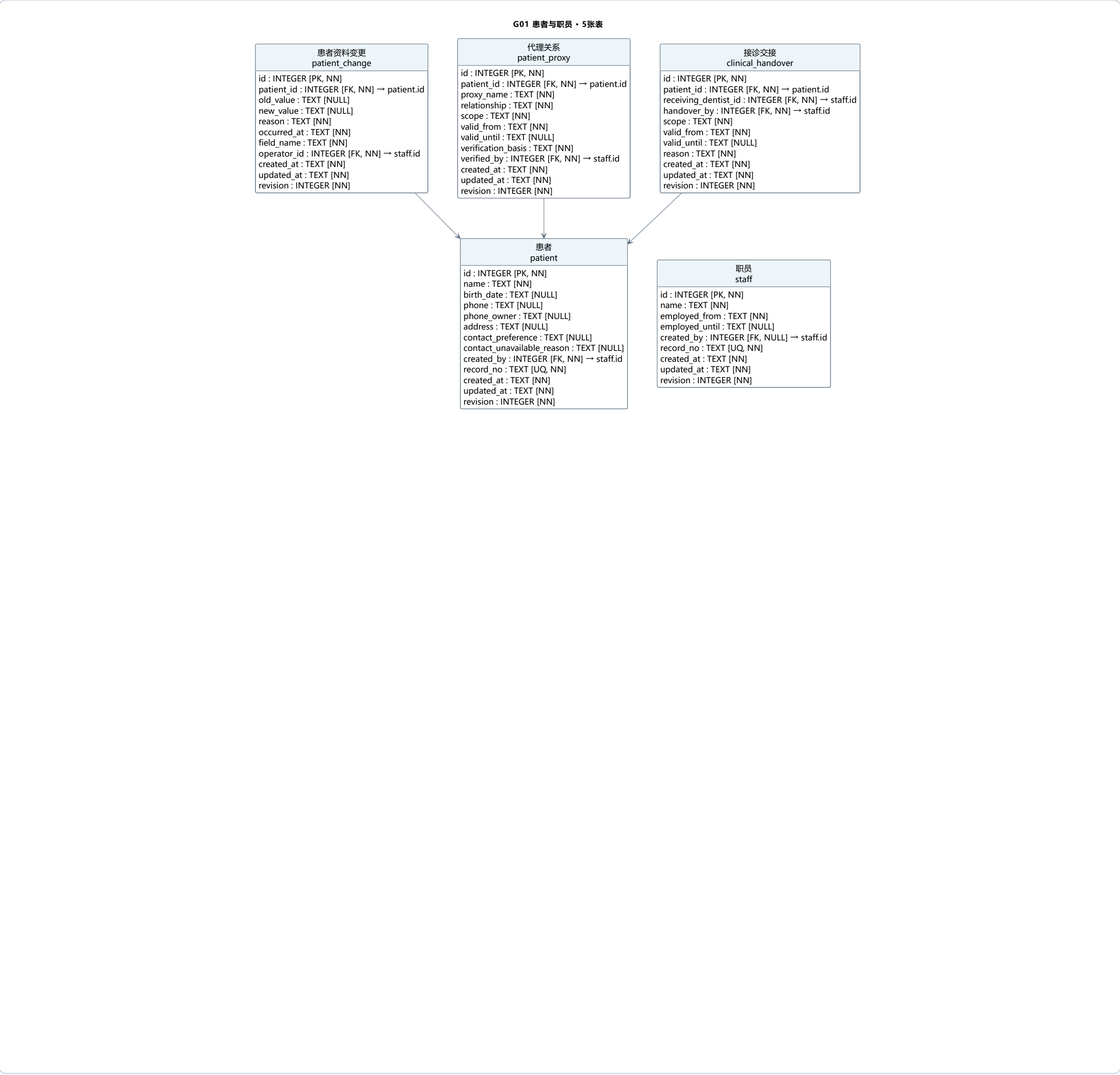
牙科诊所管理系统・数据库关系总图

85张表 / 69张业务表 + 16张运行支撑表 / 1048个字段 / SQLite目标逻辑结构

G01—G15 分区全字段视图



G01 患者与职员・5张表

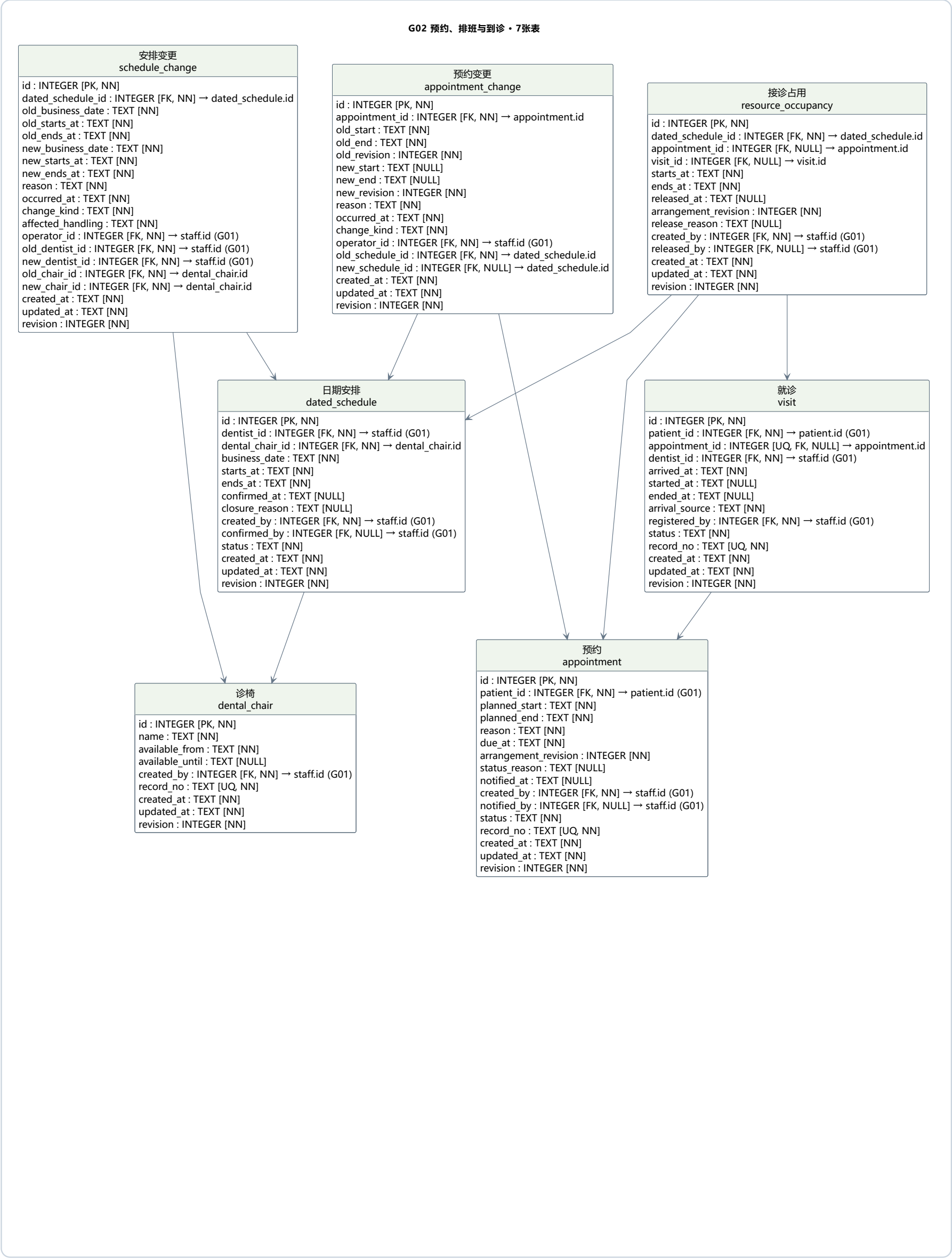
患者资料变更 patient_change
id : INTEGER [PK, NN]
patient_id : INTEGER [FK, NN] → patient.id
old_value : TEXT [NULL]
new_value : TEXT [NULL]
reason : TEXT [NN]
occurred_at : TEXT [NN]
field_name : TEXT [NN]
operator_id : INTEGER [FK, NN] → staff.id
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

代理关系 patient_proxy
id : INTEGER [PK, NN]
patient_id : INTEGER [FK, NN] → patient.id
proxy_name : TEXT [NN]
relationship : TEXT [NN]
scope : TEXT [NN]
valid_from : TEXT [NN]
valid_until : TEXT [NULL]
verification_basis : TEXT [NN]
verified_by : INTEGER [FK, NN] → staff.id
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

接诊交接 clinical_handover
id : INTEGER [PK, NN]
patient_id : INTEGER [FK, NN] → patient.id
receiving_dentist_id : INTEGER [FK, NN] → staff.id
handover_by : INTEGER [FK, NN] → staff.id
scope : TEXT [NN]
valid_from : TEXT [NN]
valid_until : TEXT [NULL]
reason : TEXT [NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

患者 patient
id : INTEGER [PK, NN]
name : TEXT [NN]
birth_date : TEXT [NULL]
phone : TEXT [NULL]
phone_owner : TEXT [NULL]
address : TEXT [NULL]
contact_preference : TEXT [NULL]
contact_unavailable_reason : TEXT [NULL]
created_by : INTEGER [FK, NN] → staff.id
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

职员 staff
id : INTEGER [PK, NN]
name : TEXT [NN]
employed_from : TEXT [NN]
employed_until : TEXT [NULL]
created_by : INTEGER [FK, NULL] → staff.id
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]



G02 预约、排班与到诊・7张表

安排变更 schedule_change
id : INTEGER [PK, NN]
dated_schedule_id : INTEGER [FK, NN] → dated_schedule.id
old_business_date : TEXT [NN]
old_starts_at : TEXT [NN]
old_ends_at : TEXT [NN]
new_business_date : TEXT [NN]
new_starts_at : TEXT [NN]
new_ends_at : TEXT [NN]
reason : TEXT [NN]
occurred_at : TEXT [NN]
change_kind : TEXT [NN]
affected_handling : TEXT [NN]
operator_id : INTEGER [FK, NN] → staff.id (G01)
old_dentist_id : INTEGER [FK, NN] → staff.id (G01)
new_dentist_id : INTEGER [FK, NN] → staff.id (G01)
old_chair_id : INTEGER [FK, NN] → dental_chair.id
new_chair_id : INTEGER [FK, NN] → dental_chair.id
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

预约变更 appointment_change
id : INTEGER [PK, NN]
appointment_id : INTEGER [FK, NN] → appointment.id
old_start : TEXT [NN]
old_end : TEXT [NN]
old_revision : INTEGER [NN]
new_start : TEXT [NULL]
new_end : TEXT [NULL]
new_revision : INTEGER [NN]
reason : TEXT [NN]
occurred_at : TEXT [NN]
change_kind : TEXT [NN]
operator_id : INTEGER [FK, NN] → staff.id (G01)
old_schedule_id : INTEGER [FK, NN] → dated_schedule.id
new_schedule_id : INTEGER [FK, NULL] → dated_schedule.id
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

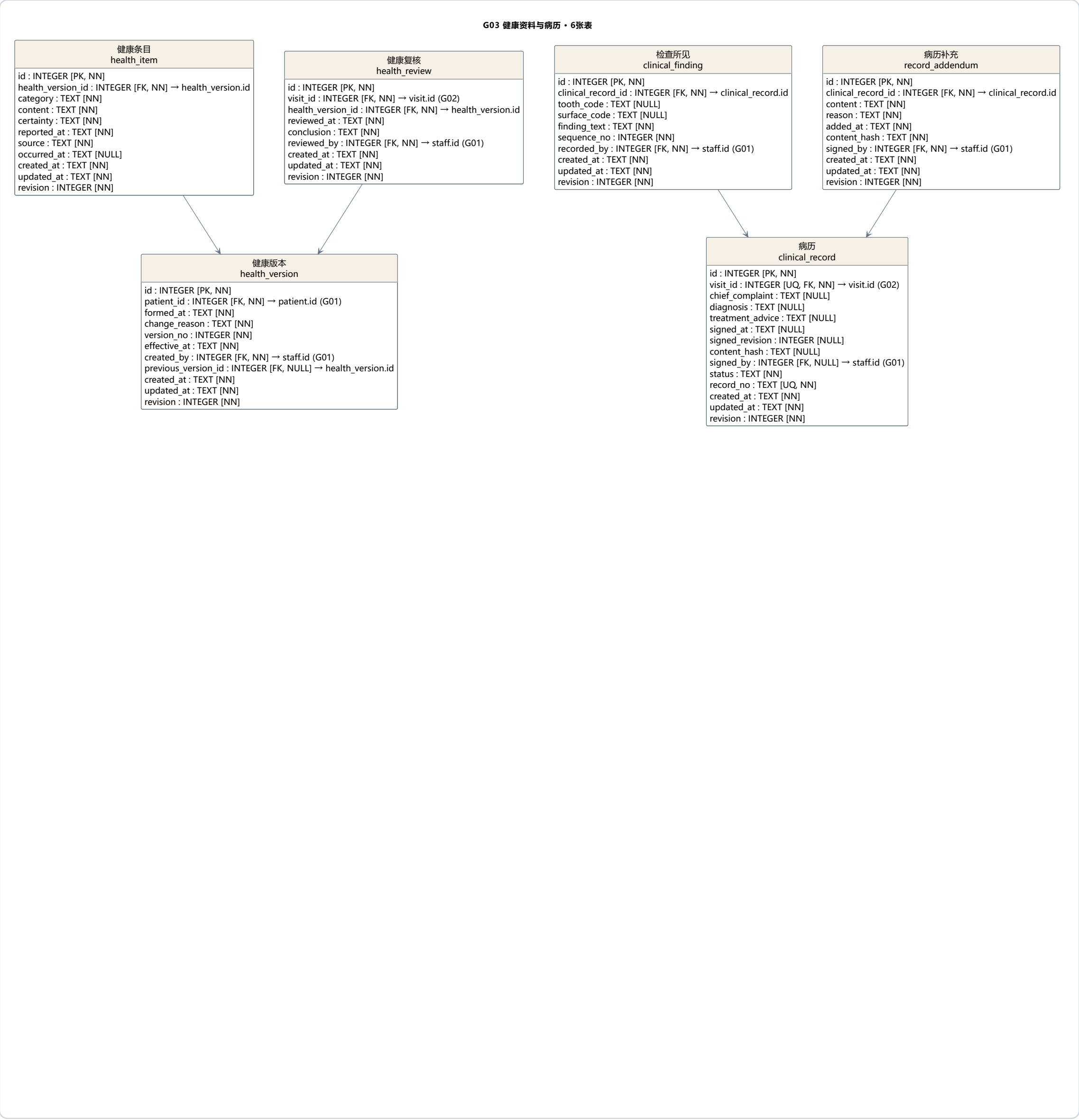
接诊占用 resource_occupancy
id : INTEGER [PK, NN]
dated_schedule_id : INTEGER [FK, NN] → dated_schedule.id
appointment_id : INTEGER [FK, NULL] → appointment.id
visit_id : INTEGER [FK, NULL] → visit.id
starts_at : TEXT [NN]
ends_at : TEXT [NN]
released_at : TEXT [NULL]
arrangement_revision : INTEGER [NN]
release_reason : TEXT [NULL]
created_by : INTEGER [FK, NN] → staff.id (G01)
released_by : INTEGER [FK, NULL] → staff.id (G01)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

日期安排 dated_schedule
id : INTEGER [PK, NN]
dentist_id : INTEGER [FK, NN] → staff.id (G01)
dental_chair_id : INTEGER [FK, NN] → dental_chair.id
business_date : TEXT [NN]
starts_at : TEXT [NN]
ends_at : TEXT [NN]
confirmed_at : TEXT [NULL]
closure_reason : TEXT [NULL]
created_by : INTEGER [FK, NN] → staff.id (G01)
confirmed_by : INTEGER [FK, NULL] → staff.id (G01)
status : TEXT [NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

就诊 visit
id : INTEGER [PK, NN]
patient_id : INTEGER [FK, NN] → patient.id (G01)
appointment_id : INTEGER [UQ, FK, NULL] → appointment.id
dentist_id : INTEGER [FK, NN] → staff.id (G01)
arrived_at : TEXT [NN]
started_at : TEXT [NULL]
ended_at : TEXT [NULL]
arrival_source : TEXT [NN]
registered_by : INTEGER [FK, NN] → staff.id (G01)
status : TEXT [NN]
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

诊椅 dental_chair
id : INTEGER [PK, NN]
name : TEXT [NN]
available_from : TEXT [NN]
available_until : TEXT [NULL]
created_by : INTEGER [FK, NN] → staff.id (G01)
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

预约 appointment
id : INTEGER [PK, NN]
patient_id : INTEGER [FK, NN] → patient.id (G01)
planned_start : TEXT [NN]
planned_end : TEXT [NN]
reason : TEXT [NN]
due_at : TEXT [NN]
arrangement_revision : INTEGER [NN]
status_reason : TEXT [NULL]
notified_at : TEXT [NULL]
created_by : INTEGER [FK, NN] → staff.id (G01)
notified_by : INTEGER [FK, NULL] → staff.id (G01)
status : TEXT [NN]
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]



G03 健康资料与病历・6张表

健康条目 health_item
id : INTEGER [PK, NN]
health_version_id : INTEGER [FK, NN] → health_version.id
category : TEXT [NN]
content : TEXT [NN]
certainty : TEXT [NN]
reported_at : TEXT [NN]
source : TEXT [NN]
occurred_at : TEXT [NULL]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

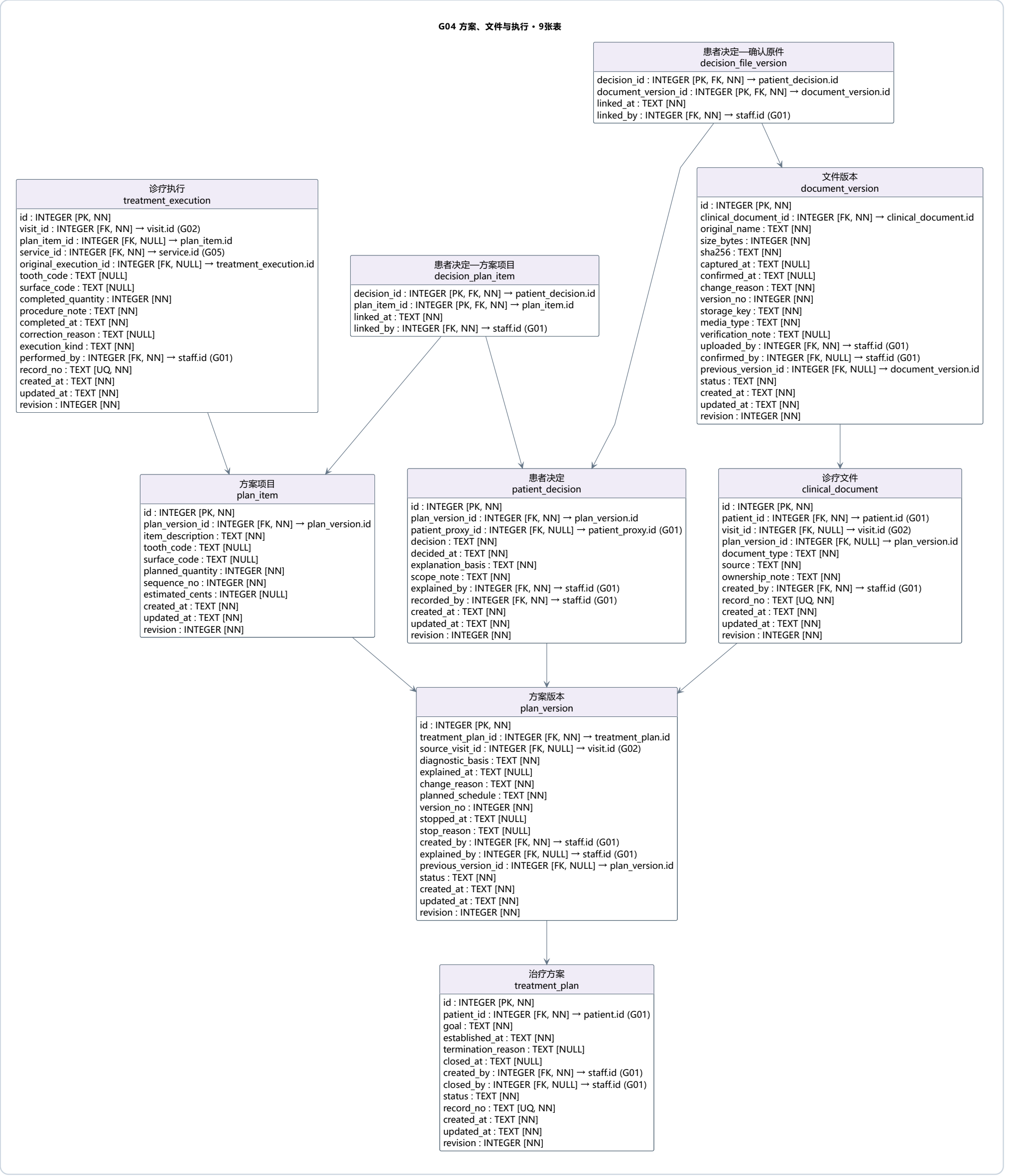
健康复核 health_review
id : INTEGER [PK, NN]
visit_id : INTEGER [FK, NN] → visit.id (G02)
health_version_id : INTEGER [FK, NN] → health_version.id
reviewed_at : TEXT [NN]
conclusion : TEXT [NN]
reviewed_by : INTEGER [FK, NN] → staff.id (G01)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

检查所见 clinical_finding
id : INTEGER [PK, NN]
clinical_record_id : INTEGER [FK, NN] → clinical_record.id
tooth_code : TEXT [NULL]
surface_code : TEXT [NULL]
finding_text : TEXT [NN]
sequence_no : INTEGER [NN]
recorded_by : INTEGER [FK, NN] → staff.id (G01)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

病历补充 record_addendum
id : INTEGER [PK, NN]
clinical_record_id : INTEGER [FK, NN] → clinical_record.id
content : TEXT [NN]
reason : TEXT [NN]
added_at : TEXT [NN]
content_hash : TEXT [NN]
signed_by : INTEGER [FK, NN] → staff.id (G01)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

健康版本 health_version
id : INTEGER [PK, NN]
patient_id : INTEGER [FK, NN] → patient.id (G01)
formed_at : TEXT [NN]
change_reason : TEXT [NN]
version_no : INTEGER [NN]
effective_at : TEXT [NN]
created_by : INTEGER [FK, NN] → staff.id (G01)
previous_version_id : INTEGER [FK, NULL] → health_version.id
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

病历 clinical_record
id : INTEGER [PK, NN]
visit_id : INTEGER [UQ, FK, NN] → visit.id (G02)
chief_complaint : TEXT [NULL]
diagnosis : TEXT [NULL]
treatment_advice : TEXT [NULL]
signed_at : TEXT [NULL]
signed_revision : INTEGER [NULL]
content_hash : TEXT [NULL]
signed_by : INTEGER [FK, NULL] → staff.id (G01)
status : TEXT [NN]
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]



G04 方案、文件与执行・9张表

患者决定—确认原件 decision_file_version
decision_id : INTEGER [PK, FK, NN] → patient_decision.id
document_version_id : INTEGER [PK, FK, NN] → document_version.id
linked_at : TEXT [NN]
linked_by : INTEGER [FK, NN] → staff.id (G01)

诊疗执行 treatment_execution
id : INTEGER [PK, NN]
visit_id : INTEGER [FK, NN] → visit.id (G02)
plan_item_id : INTEGER [FK, NULL] → plan_item.id
service_id : INTEGER [FK, NN] → service.id (G05)
original_execution_id : INTEGER [FK, NULL] → treatment_execution.id
tooth_code : TEXT [NULL]
surface_code : TEXT [NULL]
completed_quantity : INTEGER [NN]
procedure_note : TEXT [NN]
completed_at : TEXT [NN]
correction_reason : TEXT [NULL]
execution_kind : TEXT [NN]
performed_by : INTEGER [FK, NN] → staff.id (G01)
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

患者决定—方案项目 decision_plan_item
decision_id : INTEGER [PK, FK, NN] → patient_decision.id
plan_item_id : INTEGER [PK, FK, NN] → plan_item.id
linked_at : TEXT [NN]
linked_by : INTEGER [FK, NN] → staff.id (G01)

文件版本 document_version
id : INTEGER [PK, NN]
clinical_document_id : INTEGER [FK, NN] → clinical_document.id
original_name : TEXT [NN]
size_bytes : INTEGER [NN]
sha256 : TEXT [NN]
captured_at : TEXT [NULL]
confirmed_at : TEXT [NULL]
change_reason : TEXT [NN]
version_no : INTEGER [NN]
storage_key : TEXT [NN]
media_type : TEXT [NN]
verification_note : TEXT [NULL]
uploaded_by : INTEGER [FK, NN] → staff.id (G01)
confirmed_by : INTEGER [FK, NULL] → staff.id (G01)
previous_version_id : INTEGER [FK, NULL] → document_version.id
status : TEXT [NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

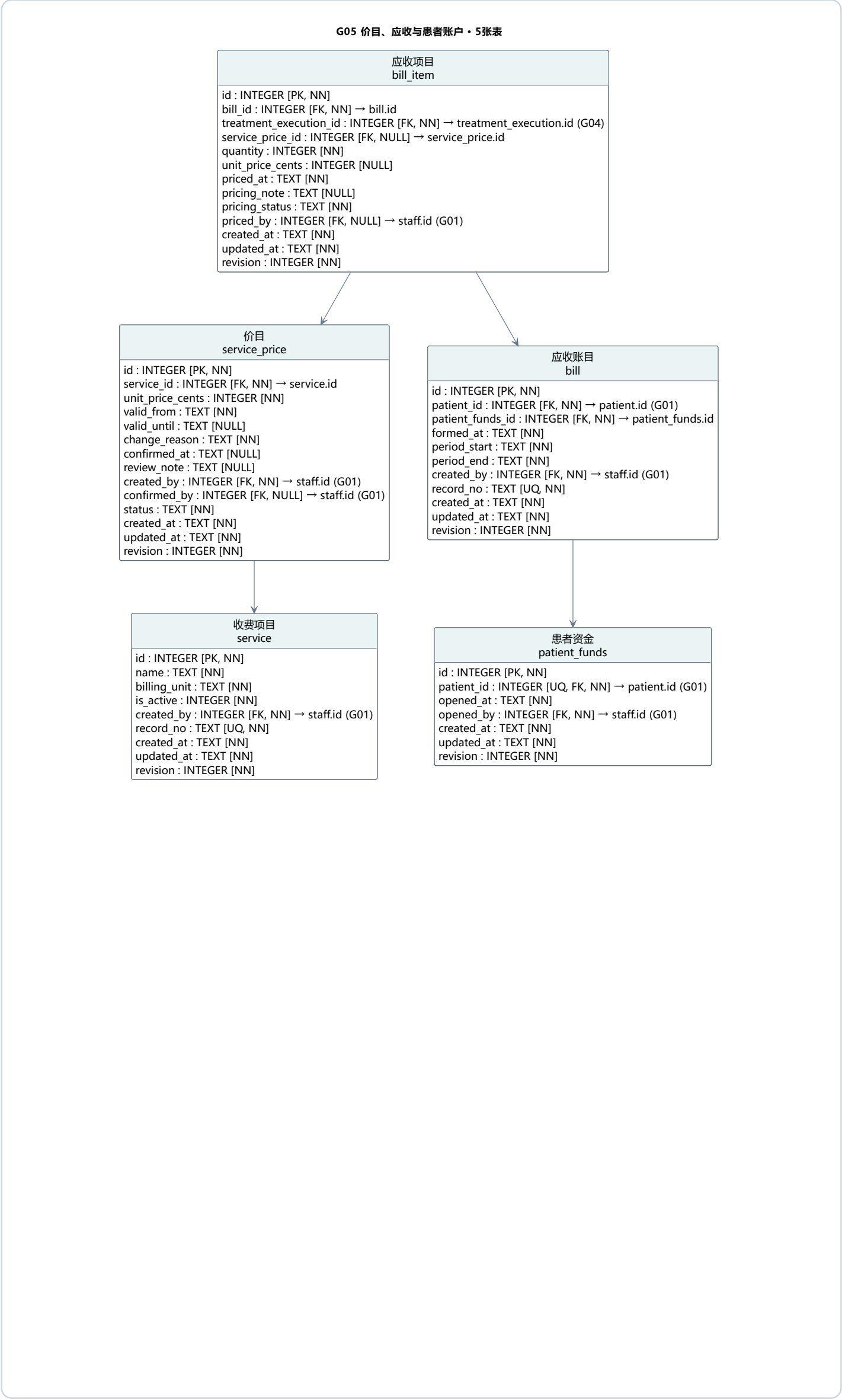
方案项目 plan_item
id : INTEGER [PK, NN]
plan_version_id : INTEGER [FK, NN] → plan_version.id
item_description : TEXT [NN]
tooth_code : TEXT [NULL]
surface_code : TEXT [NULL]
planned_quantity : INTEGER [NN]
sequence_no : INTEGER [NN]
estimated_cents : INTEGER [NULL]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

患者决定 patient_decision
id : INTEGER [PK, NN]
plan_version_id : INTEGER [FK, NN] → plan_version.id
patient_proxy_id : INTEGER [FK, NULL] → patient_proxy.id (G01)
decision : TEXT [NN]
decided_at : TEXT [NN]
explanation_basis : TEXT [NN]
scope_note : TEXT [NN]
explained_by : INTEGER [FK, NN] → staff.id (G01)
recorded_by : INTEGER [FK, NN] → staff.id (G01)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

诊疗文件 clinical_document
id : INTEGER [PK, NN]
patient_id : INTEGER [FK, NN] → patient.id (G01)
visit_id : INTEGER [FK, NULL] → visit.id (G02)
plan_version_id : INTEGER [FK, NULL] → plan_version.id
document_type : TEXT [NN]
source : TEXT [NN]
ownership_note : TEXT [NN]
created_by : INTEGER [FK, NN] → staff.id (G01)
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

方案版本 plan_version
id : INTEGER [PK, NN]
treatment_plan_id : INTEGER [FK, NN] → treatment_plan.id
source_visit_id : INTEGER [FK, NULL] → visit.id (G02)
diagnostic_basis : TEXT [NN]
explained_at : TEXT [NULL]
change_reason : TEXT [NN]
planned_schedule : TEXT [NN]
version_no : INTEGER [NN]
stopped_at : TEXT [NULL]
stop_reason : TEXT [NULL]
created_by : INTEGER [FK, NN] → staff.id (G01)
explained_by : INTEGER [FK, NULL] → staff.id (G01)
previous_version_id : INTEGER [FK, NULL] → plan_version.id
status : TEXT [NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

治疗方案 treatment_plan
id : INTEGER [PK, NN]
patient_id : INTEGER [FK, NN] → patient.id (G01)
goal : TEXT [NN]
established_at : TEXT [NN]
termination_reason : TEXT [NULL]
closed_at : TEXT [NULL]
created_by : INTEGER [FK, NN] → staff.id (G01)
closed_by : INTEGER [FK, NULL] → staff.id (G01)
status : TEXT [NN]
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]



G05 价目、应收与患者账户・5张表

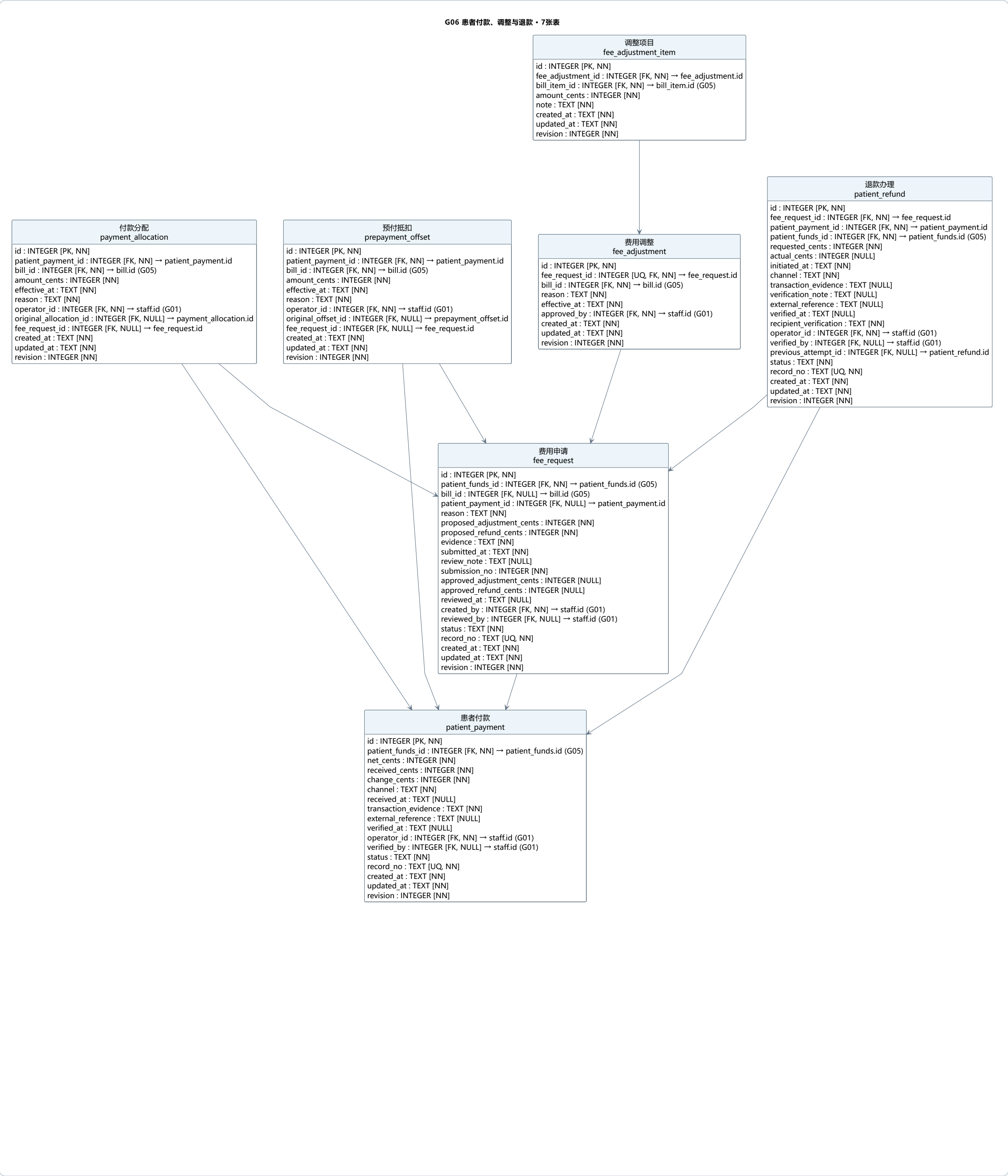
应收项目 bill_item
id : INTEGER [PK, NN]
bill_id : INTEGER [FK, NN] → bill.id
treatment_execution_id : INTEGER [FK, NN] → treatment_execution.id (G04)
service_price_id : INTEGER [FK, NULL] → service_price.id
quantity : INTEGER [NN]
unit_price_cents : INTEGER [NULL]
priced_at : TEXT [NN]
pricing_note : TEXT [NULL]
pricing_status : TEXT [NN]
priced_by : INTEGER [FK, NULL] → staff.id (G01)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

价目 service_price
id : INTEGER [PK, NN]
service_id : INTEGER [FK, NN] → service.id
unit_price_cents : INTEGER [NN]
valid_from : TEXT [NN]
valid_until : TEXT [NULL]
change_reason : TEXT [NN]
confirmed_at : TEXT [NULL]
review_note : TEXT [NULL]
created_by : INTEGER [FK, NN] → staff.id (G01)
confirmed_by : INTEGER [FK, NULL] → staff.id (G01)
status : TEXT [NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

应收账目 bill
id : INTEGER [PK, NN]
patient_id : INTEGER [FK, NN] → patient.id (G01)
patient_funds_id : INTEGER [FK, NN] → patient_funds.id
formed_at : TEXT [NN]
period_start : TEXT [NN]
period_end : TEXT [NN]
created_by : INTEGER [FK, NN] → staff.id (G01)
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

收费项目 service
id : INTEGER [PK, NN]
name : TEXT [NN]
billing_unit : TEXT [NN]
is_active : INTEGER [NN]
created_by : INTEGER [FK, NN] → staff.id (G01)
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

患者资金 patient_funds
id : INTEGER [PK, NN]
patient_id : INTEGER [UQ, FK, NN] → patient.id (G01)
opened_at : TEXT [NN]
opened_by : INTEGER [FK, NN] → staff.id (G01)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]



G06 患者付款、调整与退款・7张表

调整项目 fee_adjustment_item
id : INTEGER [PK, NN]
fee_adjustment_id : INTEGER [FK, NN] → fee_adjustment.id
bill_item_id : INTEGER [FK, NN] → bill_item.id (G05)
amount_cents : INTEGER [NN]
note : TEXT [NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

付款分配 payment_allocation
id : INTEGER [PK, NN]
patient_payment_id : INTEGER [FK, NN] → patient_payment.id
bill_id : INTEGER [FK, NN] → bill.id (G05)
amount_cents : INTEGER [NN]
effective_at : TEXT [NN]
reason : TEXT [NN]
operator_id : INTEGER [FK, NN] → staff.id (G01)
original_allocation_id : INTEGER [FK, NULL] → payment_allocation.id
fee_request_id : INTEGER [FK, NULL] → fee_request.id
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

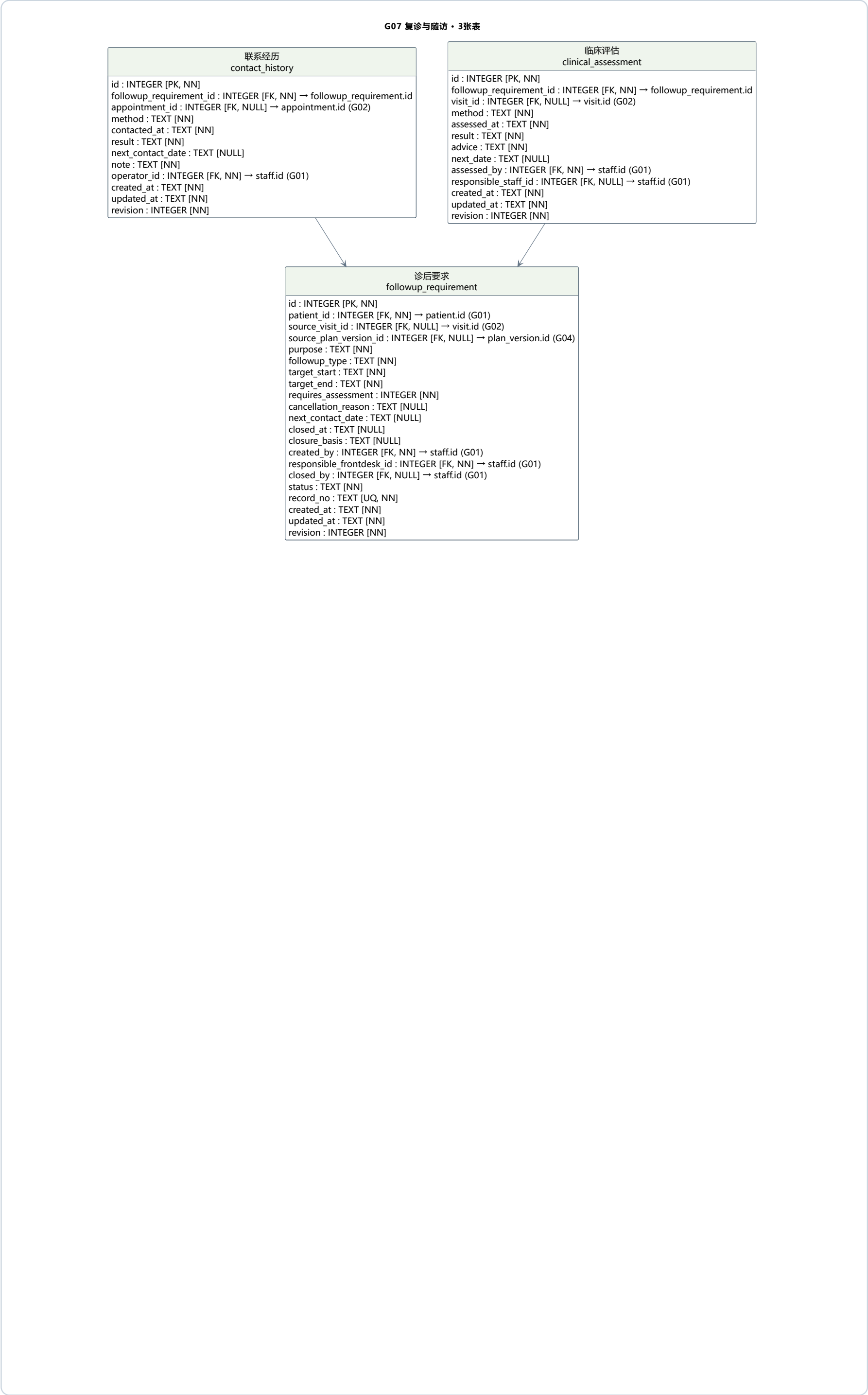
预付抵扣 prepayment_offset
id : INTEGER [PK, NN]
patient_payment_id : INTEGER [FK, NN] → patient_payment.id
bill_id : INTEGER [FK, NN] → bill.id (G05)
amount_cents : INTEGER [NN]
effective_at : TEXT [NN]
reason : TEXT [NN]
operator_id : INTEGER [FK, NN] → staff.id (G01)
original_offset_id : INTEGER [FK, NULL] → prepayment_offset.id
fee_request_id : INTEGER [FK, NULL] → fee_request.id
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

费用调整 fee_adjustment
id : INTEGER [PK, NN]
fee_request_id : INTEGER [UQ, FK, NN] → fee_request.id
bill_id : INTEGER [FK, NN] → bill.id (G05)
reason : TEXT [NN]
effective_at : TEXT [NN]
approved_by : INTEGER [FK, NN] → staff.id (G01)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

退款办理 patient_refund
id : INTEGER [PK, NN]
fee_request_id : INTEGER [FK, NN] → fee_request.id
patient_payment_id : INTEGER [FK, NN] → patient_payment.id
patient_funds_id : INTEGER [FK, NN] → patient_funds.id (G05)
requested_cents : INTEGER [NN]
actual_cents : INTEGER [NULL]
initiated_at : TEXT [NN]
channel : TEXT [NN]
transaction_evidence : TEXT [NULL]
verification_note : TEXT [NULL]
external_reference : TEXT [NULL]
verified_at : TEXT [NULL]
recipient_verification : TEXT [NN]
operator_id : INTEGER [FK, NN] → staff.id (G01)
verified_by : INTEGER [FK, NULL] → staff.id (G01)
previous_attempt_id : INTEGER [FK, NULL] → patient_refund.id
status : TEXT [NN]
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

费用申请 fee_request
id : INTEGER [PK, NN]
patient_funds_id : INTEGER [FK, NN] → patient_funds.id (G05)
bill_id : INTEGER [FK, NULL] → bill.id (G05)
patient_payment_id : INTEGER [FK, NULL] → patient_payment.id
reason : TEXT [NN]
proposed_adjustment_cents : INTEGER [NN]
proposed_refund_cents : INTEGER [NN]
evidence : TEXT [NN]
submitted_at : TEXT [NN]
review_note : TEXT [NULL]
submission_no : INTEGER [NN]
approved_adjustment_cents : INTEGER [NULL]
approved_refund_cents : INTEGER [NULL]
reviewed_at : TEXT [NULL]
created_by : INTEGER [FK, NN] → staff.id (G01)
reviewed_by : INTEGER [FK, NULL] → staff.id (G01)
status : TEXT [NN]
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

患者付款 patient_payment
id : INTEGER [PK, NN]
patient_funds_id : INTEGER [FK, NN] → patient_funds.id (G05)
net_cents : INTEGER [NN]
received_cents : INTEGER [NN]
change_cents : INTEGER [NN]
channel : TEXT [NN]
received_at : TEXT [NULL]
transaction_evidence : TEXT [NN]
external_reference : TEXT [NULL]
verified_at : TEXT [NULL]
operator_id : INTEGER [FK, NN] → staff.id (G01)
verified_by : INTEGER [FK, NULL] → staff.id (G01)
status : TEXT [NN]
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

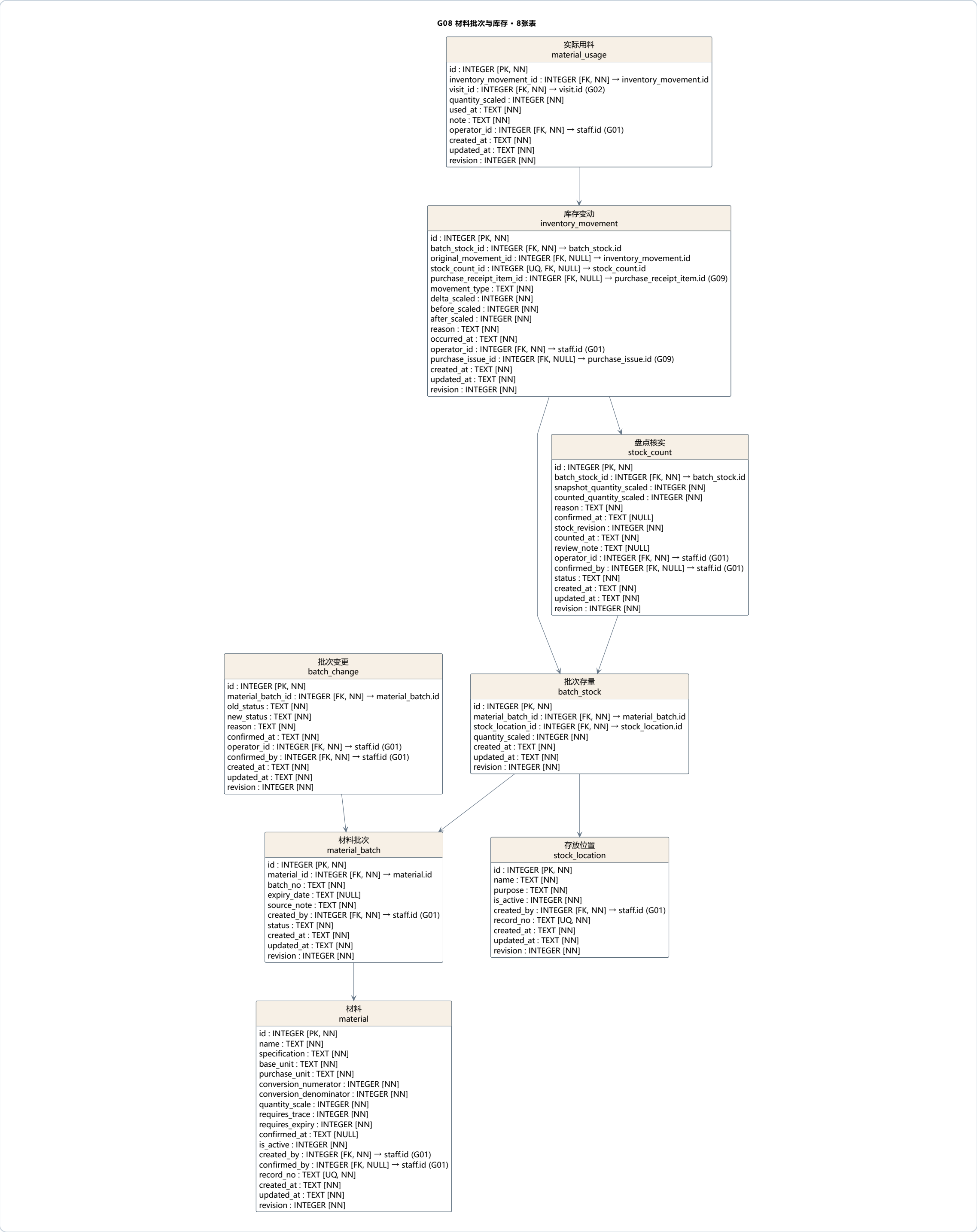


G07 复诊与随访・3张表

联系经历 contact_history
id : INTEGER [PK, NN]
followup_requirement_id : INTEGER [FK, NN] → followup_requirement.id
appointment_id : INTEGER [FK, NULL] → appointment.id (G02)
method : TEXT [NN]
contacted_at : TEXT [NN]
result : TEXT [NN]
next_contact_date : TEXT [NULL]
note : TEXT [NN]
operator_id : INTEGER [FK, NN] → staff.id (G01)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

临床评估 clinical_assessment
id : INTEGER [PK, NN]
followup_requirement_id : INTEGER [FK, NN] → followup_requirement.id
visit_id : INTEGER [FK, NULL] → visit.id (G02)
method : TEXT [NN]
assessed_at : TEXT [NN]
result : TEXT [NN]
advice : TEXT [NN]
next_date : TEXT [NULL]
assessed_by : INTEGER [FK, NN] → staff.id (G01)
responsible_staff_id : INTEGER [FK, NULL] → staff.id (G01)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

诊后要求 followup_requirement
id : INTEGER [PK, NN]
patient_id : INTEGER [FK, NN] → patient.id (G01)
source_visit_id : INTEGER [FK, NULL] → visit.id (G02)
source_plan_version_id : INTEGER [FK, NULL] → plan_version.id (G04)
purpose : TEXT [NN]
followup_type : TEXT [NN]
target_start : TEXT [NN]
target_end : TEXT [NN]
requires_assessment : INTEGER [NN]
cancellation_reason : TEXT [NULL]
next_contact_date : TEXT [NULL]
closed_at : TEXT [NULL]
closure_basis : TEXT [NULL]
created_by : INTEGER [FK, NN] → staff.id (G01)
responsible_frontdesk_id : INTEGER [FK, NN] → staff.id (G01)
closed_by : INTEGER [FK, NULL] → staff.id (G01)
status : TEXT [NN]
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]



G08 材料批次与库存・8张表

实际用料 material_usage
id : INTEGER [PK, NN]
inventory_movement_id : INTEGER [FK, NN] → inventory_movement.id
visit_id : INTEGER [FK, NN] → visit.id (G02)
quantity_scaled : INTEGER [NN]
used_at : TEXT [NN]
note : TEXT [NN]
operator_id : INTEGER [FK, NN] → staff.id (G01)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

库存变动 inventory_movement
id : INTEGER [PK, NN]
batch_stock_id : INTEGER [FK, NN] → batch_stock.id
original_movement_id : INTEGER [FK, NULL] → inventory_movement.id
stock_count_id : INTEGER [UQ, FK, NULL] → stock_count.id
purchase_receipt_item_id : INTEGER [FK, NULL] → purchase_receipt_item.id (G09)
movement_type : TEXT [NN]
delta_scaled : INTEGER [NN]
before_scaled : INTEGER [NN]
after_scaled : INTEGER [NN]
reason : TEXT [NN]
occurred_at : TEXT [NN]
operator_id : INTEGER [FK, NN] → staff.id (G01)
purchase_issue_id : INTEGER [FK, NULL] → purchase_issue.id (G09)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

盘点核实 stock_count
id : INTEGER [PK, NN]
batch_stock_id : INTEGER [FK, NN] → batch_stock.id
snapshot_quantity_scaled : INTEGER [NN]
counted_quantity_scaled : INTEGER [NN]
reason : TEXT [NN]
confirmed_at : TEXT [NULL]
stock_revision : INTEGER [NN]
counted_at : TEXT [NN]
review_note : TEXT [NULL]
operator_id : INTEGER [FK, NN] → staff.id (G01)
confirmed_by : INTEGER [FK, NULL] → staff.id (G01)
status : TEXT [NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

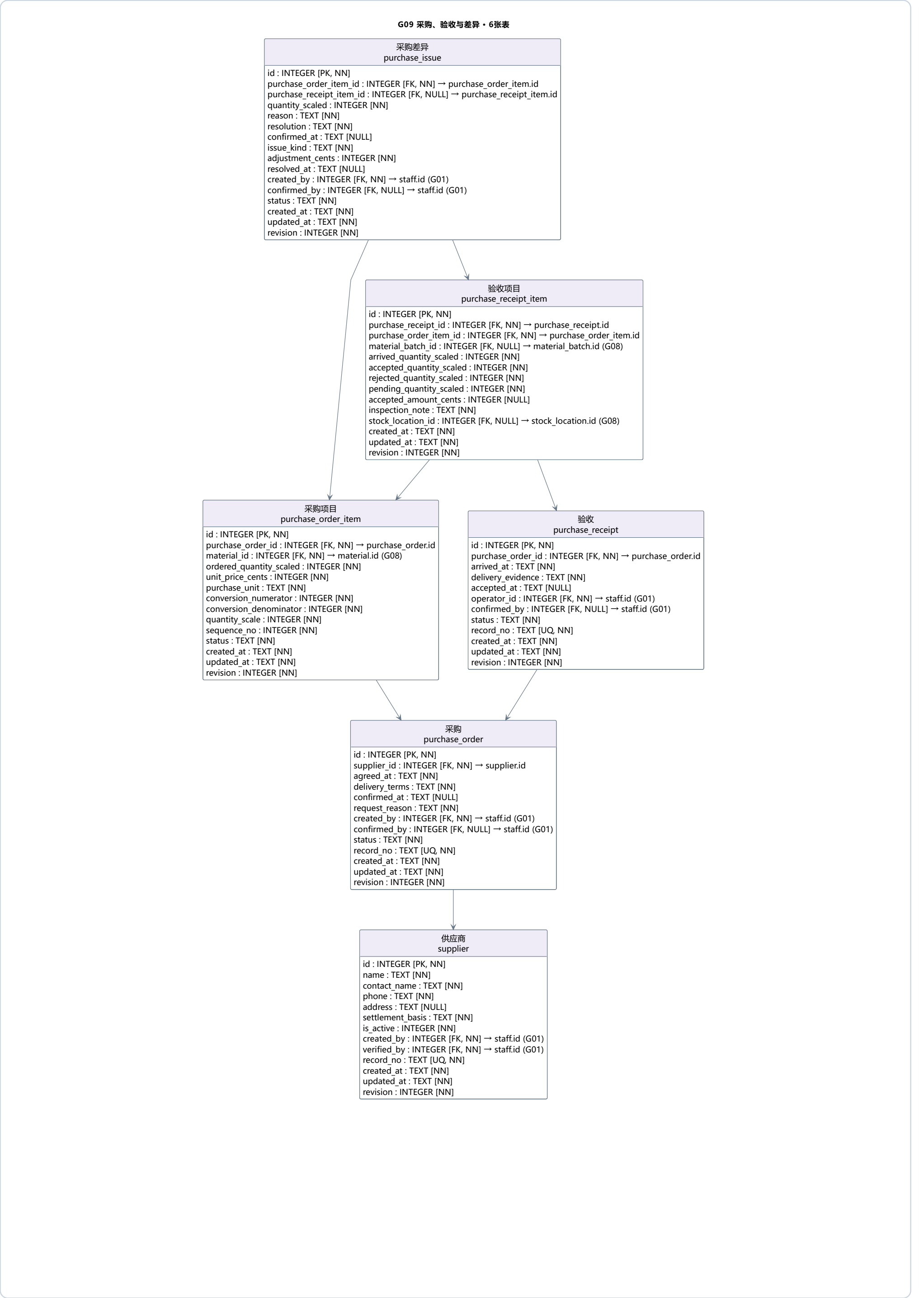
批次变更 batch_change
id : INTEGER [PK, NN]
material_batch_id : INTEGER [FK, NN] → material_batch.id
old_status : TEXT [NN]
new_status : TEXT [NN]
reason : TEXT [NN]
confirmed_at : TEXT [NN]
operator_id : INTEGER [FK, NN] → staff.id (G01)
confirmed_by : INTEGER [FK, NN] → staff.id (G01)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

批次存量 batch_stock
id : INTEGER [PK, NN]
material_batch_id : INTEGER [FK, NN] → material_batch.id
stock_location_id : INTEGER [FK, NN] → stock_location.id
quantity_scaled : INTEGER [NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

材料批次 material_batch
id : INTEGER [PK, NN]
material_id : INTEGER [FK, NN] → material.id
batch_no : TEXT [NN]
expiry_date : TEXT [NULL]
source_note : TEXT [NN]
created_by : INTEGER [FK, NN] → staff.id (G01)
status : TEXT [NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

存放位置 stock_location
id : INTEGER [PK, NN]
name : TEXT [NN]
purpose : TEXT [NN]
is_active : INTEGER [NN]
created_by : INTEGER [FK, NN] → staff.id (G01)
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

材料 material
id : INTEGER [PK, NN]
name : TEXT [NN]
specification : TEXT [NN]
base_unit : TEXT [NN]
purchase_unit : TEXT [NN]
conversion_numerator : INTEGER [NN]
conversion_denominator : INTEGER [NN]
quantity_scale : INTEGER [NN]
requires_trace : INTEGER [NN]
requires_expiry : INTEGER [NN]
confirmed_at : TEXT [NULL]
is_active : INTEGER [NN]
created_by : INTEGER [FK, NN] → staff.id (G01)
confirmed_by : INTEGER [FK, NULL] → staff.id (G01)
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]



G09 采购、验收与差异・6张表

采购差异 purchase_issue
id : INTEGER [PK, NN]
purchase_order_item_id : INTEGER [FK, NN] → purchase_order_item.id
purchase_receipt_item_id : INTEGER [FK, NULL] → purchase_receipt_item.id
quantity_scaled : INTEGER [NN]
reason : TEXT [NN]
resolution : TEXT [NN]
confirmed_at : TEXT [NULL]
issue_kind : TEXT [NN]
adjustment_cents : INTEGER [NN]
resolved_at : TEXT [NULL]
created_by : INTEGER [FK, NN] → staff.id (G01)
confirmed_by : INTEGER [FK, NULL] → staff.id (G01)
status : TEXT [NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

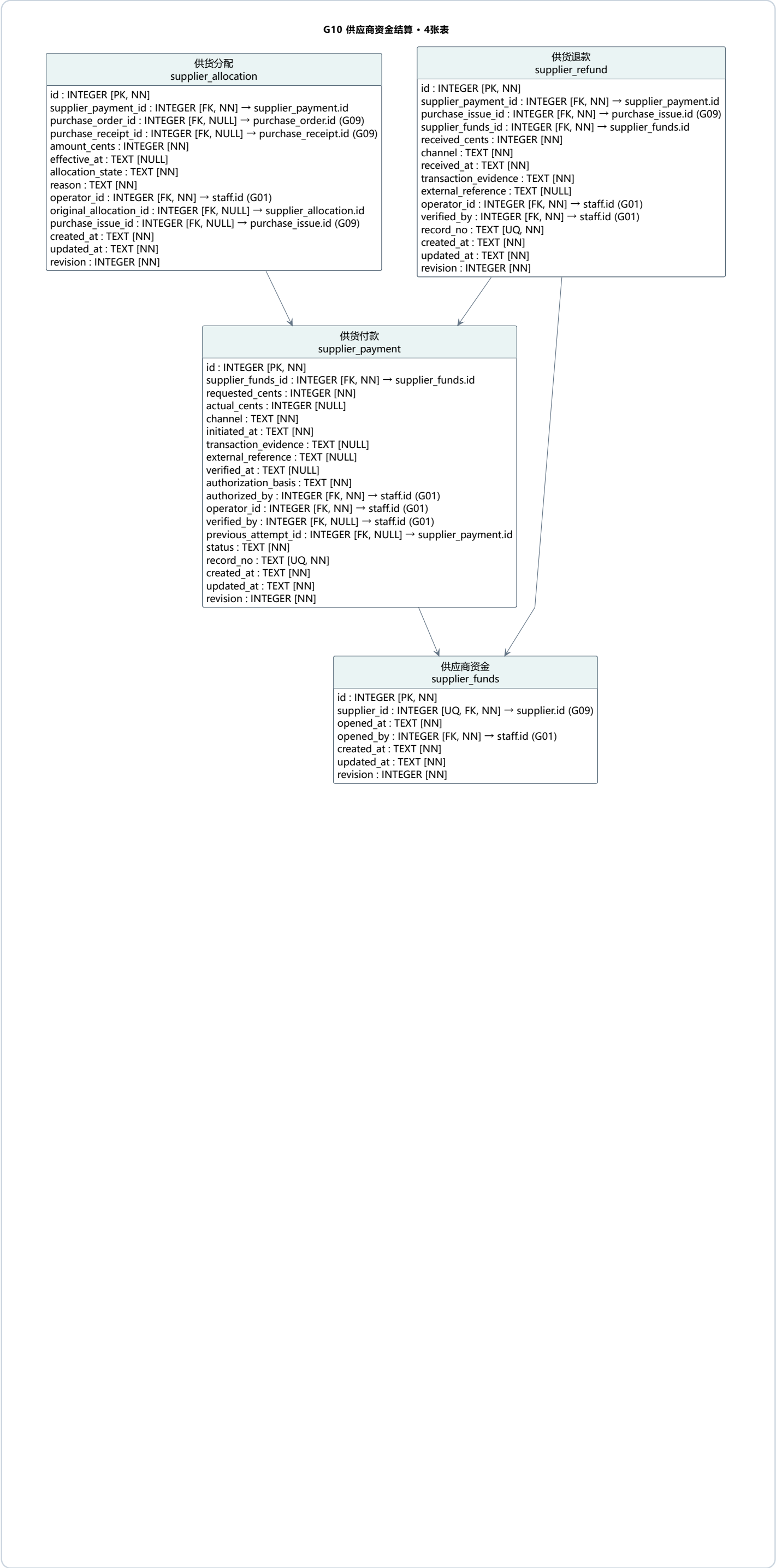
验收项目 purchase_receipt_item
id : INTEGER [PK, NN]
purchase_receipt_id : INTEGER [FK, NN] → purchase_receipt.id
purchase_order_item_id : INTEGER [FK, NN] → purchase_order_item.id
material_batch_id : INTEGER [FK, NULL] → material_batch.id (G08)
arrived_quantity_scaled : INTEGER [NN]
accepted_quantity_scaled : INTEGER [NN]
rejected_quantity_scaled : INTEGER [NN]
pending_quantity_scaled : INTEGER [NN]
accepted_amount_cents : INTEGER [NULL]
inspection_note : TEXT [NN]
stock_location_id : INTEGER [FK, NULL] → stock_location.id (G08)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

采购项目 purchase_order_item
id : INTEGER [PK, NN]
purchase_order_id : INTEGER [FK, NN] → purchase_order.id
material_id : INTEGER [FK, NN] → material.id (G08)
ordered_quantity_scaled : INTEGER [NN]
unit_price_cents : INTEGER [NN]
purchase_unit : TEXT [NN]
conversion_numerator : INTEGER [NN]
conversion_denominator : INTEGER [NN]
quantity_scale : INTEGER [NN]
sequence_no : INTEGER [NN]
status : TEXT [NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

验收 purchase_receipt
id : INTEGER [PK, NN]
purchase_order_id : INTEGER [FK, NN] → purchase_order.id
arrived_at : TEXT [NN]
delivery_evidence : TEXT [NN]
accepted_at : TEXT [NULL]
operator_id : INTEGER [FK, NN] → staff.id (G01)
confirmed_by : INTEGER [FK, NULL] → staff.id (G01)
status : TEXT [NN]
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

采购 purchase_order
id : INTEGER [PK, NN]
supplier_id : INTEGER [FK, NN] → supplier.id
agreed_at : TEXT [NN]
delivery_terms : TEXT [NN]
confirmed_at : TEXT [NULL]
request_reason : TEXT [NN]
created_by : INTEGER [FK, NN] → staff.id (G01)
confirmed_by : INTEGER [FK, NULL] → staff.id (G01)
status : TEXT [NN]
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

供应商 supplier
id : INTEGER [PK, NN]
name : TEXT [NN]
contact_name : TEXT [NN]
phone : TEXT [NN]
address : TEXT [NULL]
settlement_basis : TEXT [NN]
is_active : INTEGER [NN]
created_by : INTEGER [FK, NN] → staff.id (G01)
verified_by : INTEGER [FK, NN] → staff.id (G01)
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]



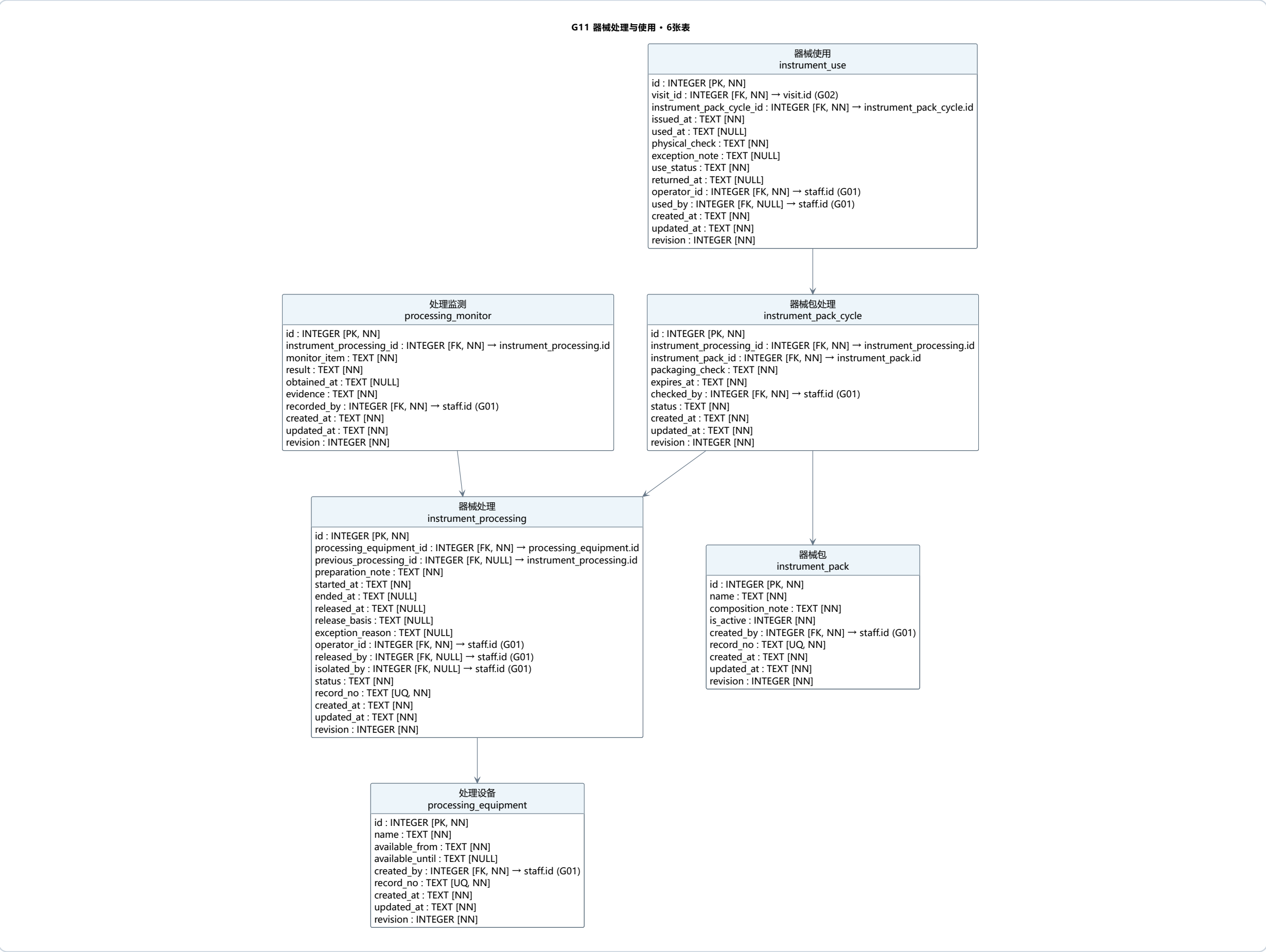
G10 供应商资金结算・4张表

供货分配 supplier_allocation
id : INTEGER [PK, NN]
supplier_payment_id : INTEGER [FK, NN] → supplier_payment.id
purchase_order_id : INTEGER [FK, NULL] → purchase_order.id (G09)
purchase_receipt_id : INTEGER [FK, NULL] → purchase_receipt.id (G09)
amount_cents : INTEGER [NN]
effective_at : TEXT [NULL]
allocation_state : TEXT [NN]
reason : TEXT [NN]
operator_id : INTEGER [FK, NN] → staff.id (G01)
original_allocation_id : INTEGER [FK, NULL] → supplier_allocation.id
purchase_issue_id : INTEGER [FK, NULL] → purchase_issue.id (G09)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

供货退款 supplier_refund
id : INTEGER [PK, NN]
supplier_payment_id : INTEGER [FK, NN] → supplier_payment.id
purchase_issue_id : INTEGER [FK, NN] → purchase_issue.id (G09)
supplier_funds_id : INTEGER [FK, NN] → supplier_funds.id
received_cents : INTEGER [NN]
channel : TEXT [NN]
received_at : TEXT [NN]
transaction_evidence : TEXT [NN]
external_reference : TEXT [NULL]
operator_id : INTEGER [FK, NN] → staff.id (G01)
verified_by : INTEGER [FK, NN] → staff.id (G01)
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

供货付款 supplier_payment
id : INTEGER [PK, NN]
supplier_funds_id : INTEGER [FK, NN] → supplier_funds.id
requested_cents : INTEGER [NN]
actual_cents : INTEGER [NULL]
channel : TEXT [NN]
initiated_at : TEXT [NN]
transaction_evidence : TEXT [NULL]
external_reference : TEXT [NULL]
verified_at : TEXT [NULL]
authorization_basis : TEXT [NN]
authorized_by : INTEGER [FK, NN] → staff.id (G01)
operator_id : INTEGER [FK, NN] → staff.id (G01)
verified_by : INTEGER [FK, NULL] → staff.id (G01)
previous_attempt_id : INTEGER [FK, NULL] → supplier_payment.id
status : TEXT [NN]
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

供应商资金 supplier_funds
id : INTEGER [PK, NN]
supplier_id : INTEGER [UQ, FK, NN] → supplier.id (G09)
opened_at : TEXT [NN]
opened_by : INTEGER [FK, NN] → staff.id (G01)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]



G11 器械处理与使用・6张表

器械使用 instrument_use
id : INTEGER [PK, NN]
visit_id : INTEGER [FK, NN] → visit.id (G02)
instrument_pack_cycle_id : INTEGER [FK, NN] → instrument_pack_cycle.id
issued_at : TEXT [NN]
used_at : TEXT [NULL]
physical_check : TEXT [NN]
exception_note : TEXT [NULL]
use_status : TEXT [NN]
returned_at : TEXT [NULL]
operator_id : INTEGER [FK, NN] → staff.id (G01)
used_by : INTEGER [FK, NULL] → staff.id (G01)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

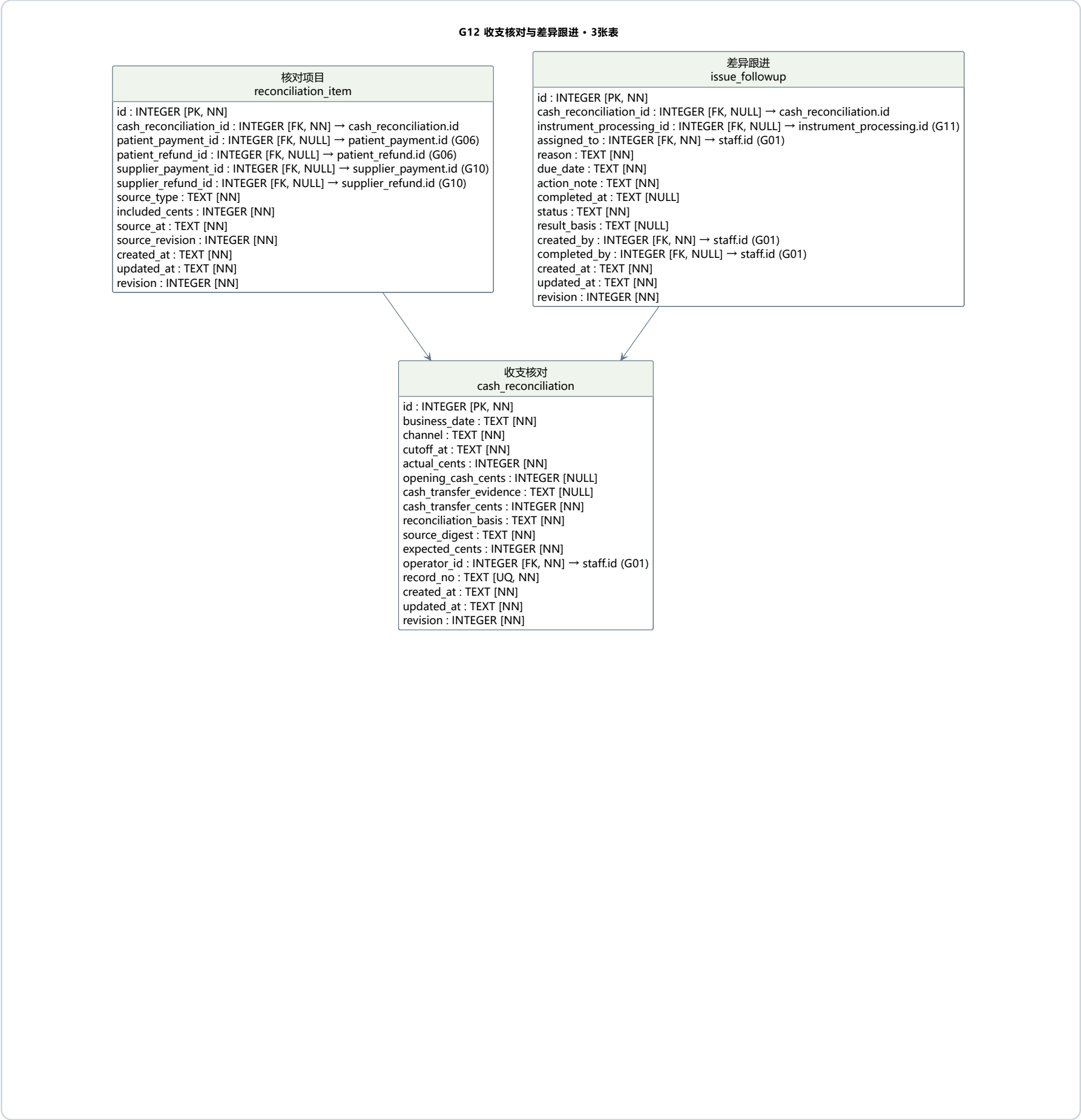
处理监测 processing_monitor
id : INTEGER [PK, NN]
instrument_processing_id : INTEGER [FK, NN] → instrument_processing.id
monitor_item : TEXT [NN]
result : TEXT [NN]
obtained_at : TEXT [NULL]
evidence : TEXT [NN]
recorded_by : INTEGER [FK, NN] → staff.id (G01)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

器械包处理 instrument_pack_cycle
id : INTEGER [PK, NN]
instrument_processing_id : INTEGER [FK, NN] → instrument_processing.id
instrument_pack_id : INTEGER [FK, NN] → instrument_pack.id
packaging_check : TEXT [NN]
expires_at : TEXT [NN]
checked_by : INTEGER [FK, NN] → staff.id (G01)
status : TEXT [NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

器械处理 instrument_processing
id : INTEGER [PK, NN]
processing_equipment_id : INTEGER [FK, NN] → processing_equipment.id
previous_processing_id : INTEGER [FK, NULL] → instrument_processing.id
preparation_note : TEXT [NN]
started_at : TEXT [NN]
ended_at : TEXT [NULL]
released_at : TEXT [NULL]
release_basis : TEXT [NULL]
exception_reason : TEXT [NULL]
operator_id : INTEGER [FK, NN] → staff.id (G01)
released_by : INTEGER [FK, NULL] → staff.id (G01)
isolated_by : INTEGER [FK, NULL] → staff.id (G01)
status : TEXT [NN]
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

器械包 instrument_pack
id : INTEGER [PK, NN]
name : TEXT [NN]
composition_note : TEXT [NN]
is_active : INTEGER [NN]
created_by : INTEGER [FK, NN] → staff.id (G01)
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

处理设备 processing_equipment
id : INTEGER [PK, NN]
name : TEXT [NN]
available_from : TEXT [NN]
available_until : TEXT [NULL]
created_by : INTEGER [FK, NN] → staff.id (G01)
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

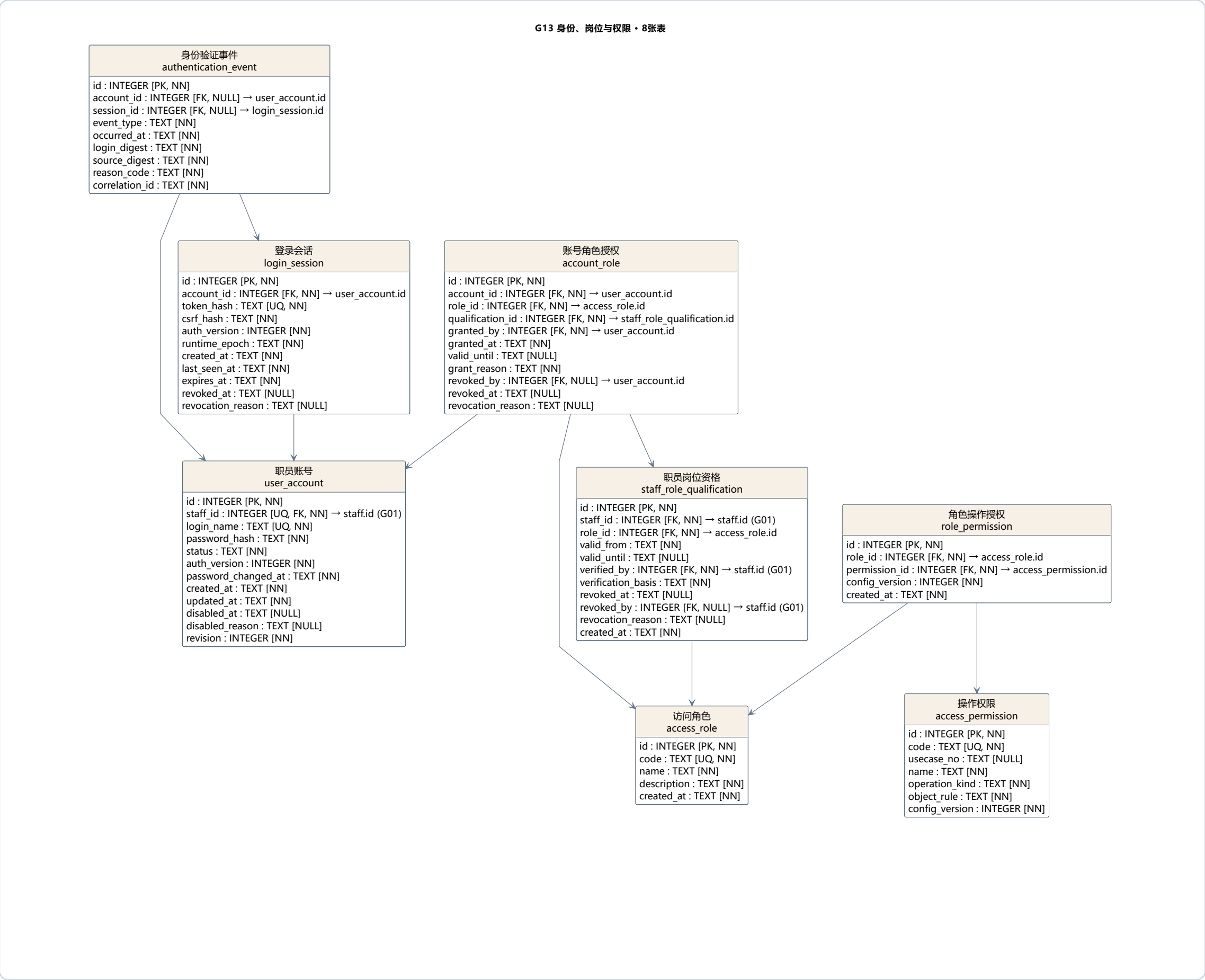


G12 收支核对与差异跟进・3张表

核对项目 reconciliation_item
id : INTEGER [PK, NN]
cash_reconciliation_id : INTEGER [FK, NN] → cash_reconciliation.id
patient_payment_id : INTEGER [FK, NULL] → patient_payment.id (G06)
patient_refund_id : INTEGER [FK, NULL] → patient_refund.id (G06)
supplier_payment_id : INTEGER [FK, NULL] → supplier_payment.id (G10)
supplier_refund_id : INTEGER [FK, NULL] → supplier_refund.id (G10)
source_type : TEXT [NN]
included_cents : INTEGER [NN]
source_at : TEXT [NN]
source_revision : INTEGER [NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

差异跟进 issue_followup
id : INTEGER [PK, NN]
cash_reconciliation_id : INTEGER [FK, NULL] → cash_reconciliation.id
instrument_processing_id : INTEGER [FK, NULL] → instrument_processing.id (G11)
assigned_to : INTEGER [FK, NN] → staff.id (G01)
reason : TEXT [NN]
due_date : TEXT [NN]
action_note : TEXT [NN]
completed_at : TEXT [NULL]
status : TEXT [NN]
result_basis : TEXT [NULL]
created_by : INTEGER [FK, NN] → staff.id (G01)
completed_by : INTEGER [FK, NULL] → staff.id (G01)
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]

收支核对 cash_reconciliation
id : INTEGER [PK, NN]
business_date : TEXT [NN]
channel : TEXT [NN]
cutoff_at : TEXT [NN]
actual_cents : INTEGER [NN]
opening_cash_cents : INTEGER [NULL]
cash_transfer_evidence : TEXT [NULL]
cash_transfer_cents : INTEGER [NN]
reconciliation_basis : TEXT [NN]
source_digest : TEXT [NN]
expected_cents : INTEGER [NN]
operator_id : INTEGER [FK, NN] → staff.id (G01)
record_no : TEXT [UQ, NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
revision : INTEGER [NN]



G13 身份、岗位与权限・8张表

身份验证事件 authentication_event
id : INTEGER [PK, NN]
account_id : INTEGER [FK, NULL] → user_account.id
session_id : INTEGER [FK, NULL] → login_session.id
event_type : TEXT [NN]
occurred_at : TEXT [NN]
login_digest : TEXT [NN]
source_digest : TEXT [NN]
reason_code : TEXT [NN]
correlation_id : TEXT [NN]

登录会话 login_session
id : INTEGER [PK, NN]
account_id : INTEGER [FK, NN] → user_account.id
token_hash : TEXT [UQ, NN]
csrf_hash : TEXT [NN]
auth_version : INTEGER [NN]
runtime_epoch : TEXT [NN]
created_at : TEXT [NN]
last_seen_at : TEXT [NN]
expires_at : TEXT [NN]
revoked_at : TEXT [NULL]
revocation_reason : TEXT [NULL]

账号角色授权 account_role
id : INTEGER [PK, NN]
account_id : INTEGER [FK, NN] → user_account.id
role_id : INTEGER [FK, NN] → access_role.id
qualification_id : INTEGER [FK, NN] → staff_role_qualification.id
granted_by : INTEGER [FK, NN] → user_account.id
granted_at : TEXT [NN]
valid_until : TEXT [NULL]
grant_reason : TEXT [NN]
revoked_by : INTEGER [FK, NULL] → user_account.id
revoked_at : TEXT [NULL]
revocation_reason : TEXT [NULL]

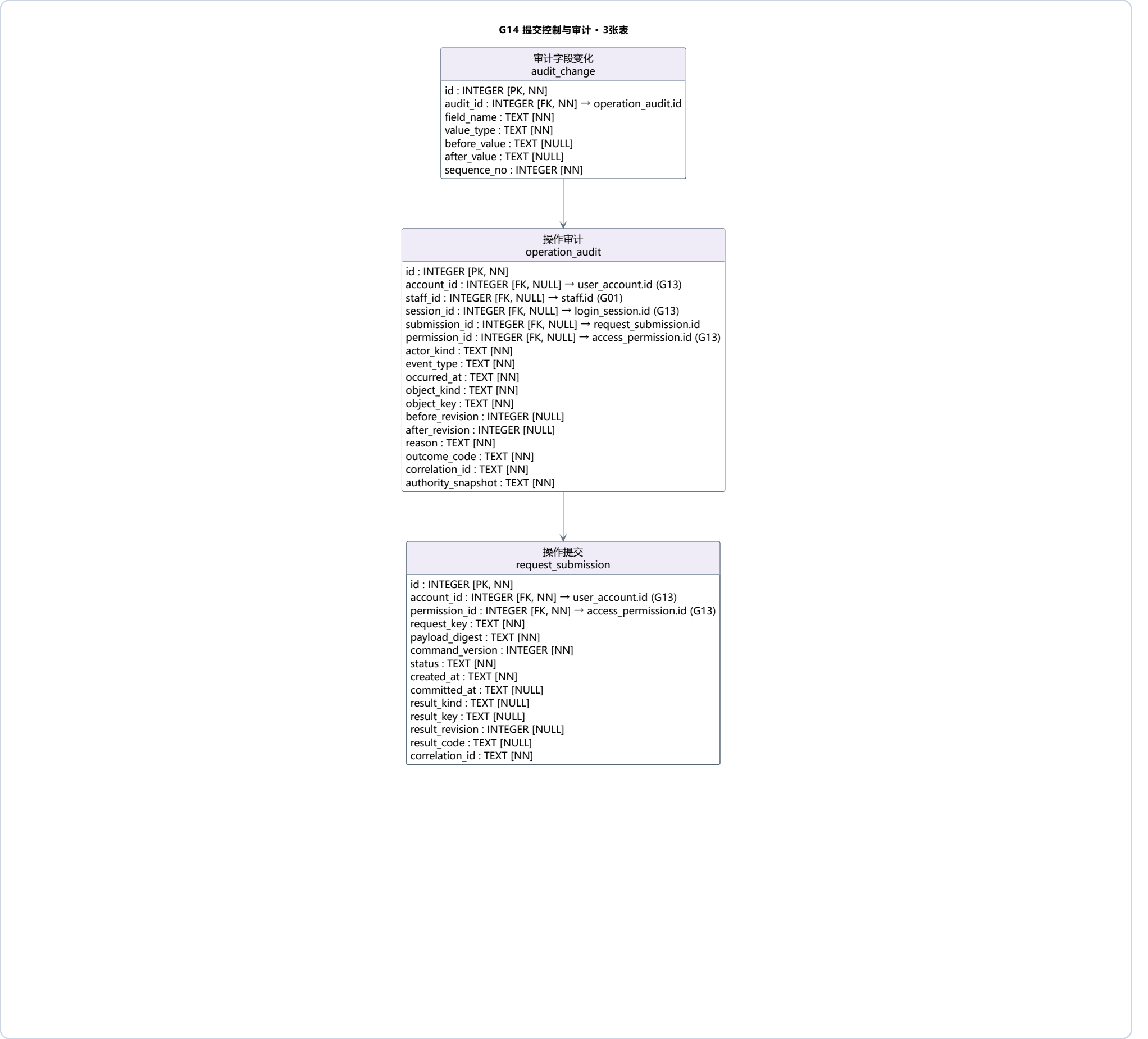
职员账号 user_account
id : INTEGER [PK, NN]
staff_id : INTEGER [UQ, FK, NN] → staff.id (G01)
login_name : TEXT [UQ, NN]
password_hash : TEXT [NN]
status : TEXT [NN]
auth_version : INTEGER [NN]
password_changed_at : TEXT [NN]
created_at : TEXT [NN]
updated_at : TEXT [NN]
disabled_at : TEXT [NULL]
disabled_reason : TEXT [NULL]
revision : INTEGER [NN]

职员岗位资格 staff_role_qualification
id : INTEGER [PK, NN]
staff_id : INTEGER [FK, NN] → staff.id (G01)
role_id : INTEGER [FK, NN] → access_role.id
valid_from : TEXT [NN]
valid_until : TEXT [NULL]
verified_by : INTEGER [FK, NN] → staff.id (G01)
verification_basis : TEXT [NN]
revoked_at : TEXT [NULL]
revoked_by : INTEGER [FK, NULL] → staff.id (G01)
revocation_reason : TEXT [NULL]
created_at : TEXT [NN]

角色操作授权 role_permission
id : INTEGER [PK, NN]
role_id : INTEGER [FK, NN] → access_role.id
permission_id : INTEGER [FK, NN] → access_permission.id
config_version : INTEGER [NN]
created_at : TEXT [NN]

访问角色 access_role
id : INTEGER [PK, NN]
code : TEXT [UQ, NN]
name : TEXT [NN]
description : TEXT [NN]
created_at : TEXT [NN]

操作权限 access_permission
id : INTEGER [PK, NN]
code : TEXT [UQ, NN]
usecase_no : TEXT [NULL]
name : TEXT [NN]
operation_kind : TEXT [NN]
object_rule : TEXT [NN]
config_version : INTEGER [NN]

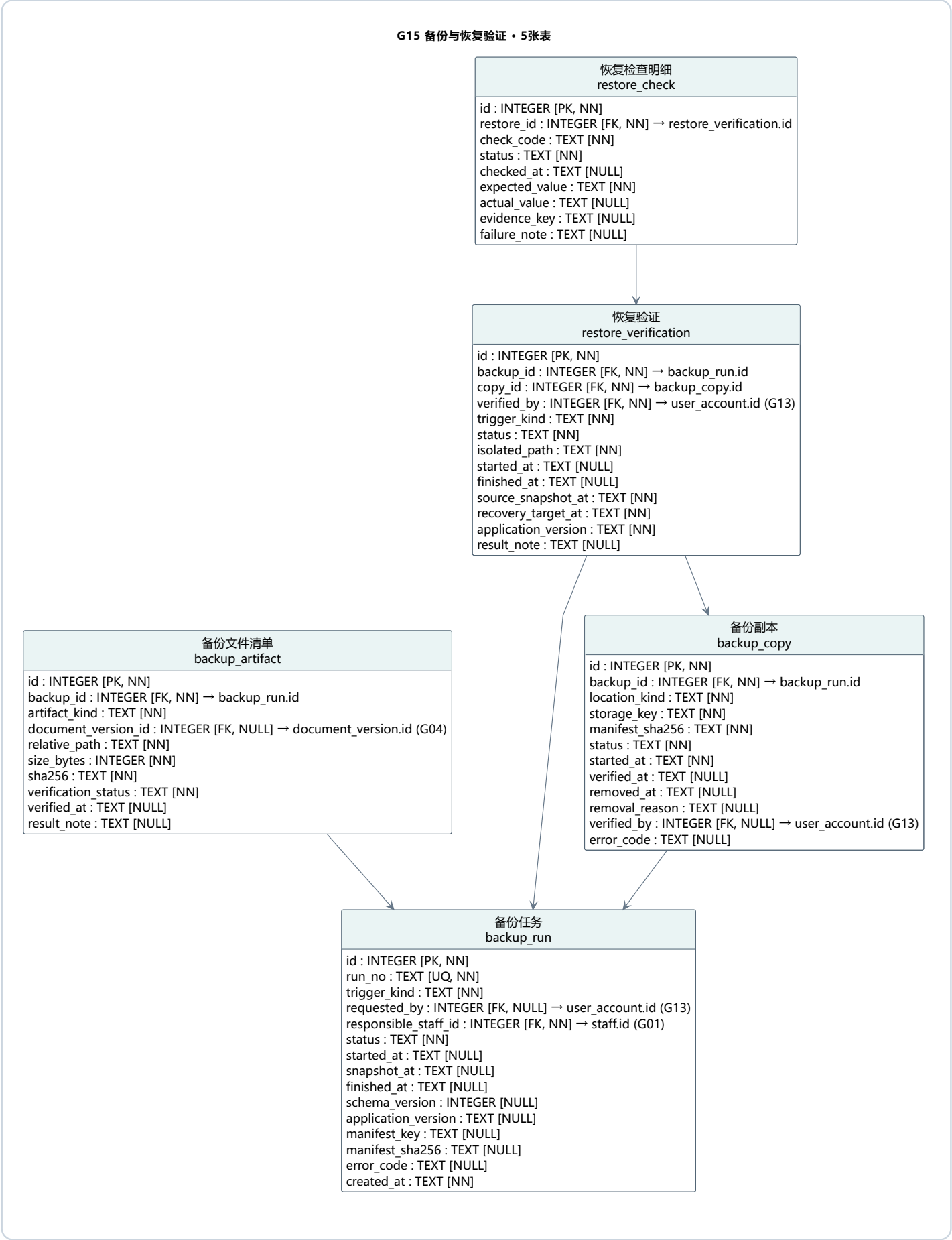


G14 提交控制与审计・3张表

审计字段变化 audit_change
id : INTEGER [PK, NN]
audit_id : INTEGER [FK, NN] → operation_audit.id
field_name : TEXT [NN]
value_type : TEXT [NN]
before_value : TEXT [NULL]
after_value : TEXT [NULL]
sequence_no : INTEGER [NN]

操作审计 operation_audit
id : INTEGER [PK, NN]
account_id : INTEGER [FK, NULL] → user_account.id (G13)
staff_id : INTEGER [FK, NULL] → staff.id (G01)
session_id : INTEGER [FK, NULL] → login_session.id (G13)
submission_id : INTEGER [FK, NULL] → request_submission.id
permission_id : INTEGER [FK, NULL] → access_permission.id (G13)
actor_kind : TEXT [NN]
event_type : TEXT [NN]
occurred_at : TEXT [NN]
object_kind : TEXT [NN]
object_key : TEXT [NN]
before_revision : INTEGER [NULL]
after_revision : INTEGER [NULL]
reason : TEXT [NN]
outcome_code : TEXT [NN]
correlation_id : TEXT [NN]
authority_snapshot : TEXT [NN]

操作提交 request_submission
id : INTEGER [PK, NN]
account_id : INTEGER [FK, NN] → user_account.id (G13)
permission_id : INTEGER [FK, NN] → access_permission.id (G13)
request_key : TEXT [NN]
payload_digest : TEXT [NN]
command_version : INTEGER [NN]
status : TEXT [NN]
created_at : TEXT [NN]
committed_at : TEXT [NULL]
result_kind : TEXT [NULL]
result_key : TEXT [NULL]
result_revision : INTEGER [NULL]
result_code : TEXT [NULL]
correlation_id : TEXT [NN]



G15 备份与恢复验证・5张表

恢复检查明细 restore_check
id : INTEGER [PK, NN]
restore_id : INTEGER [FK, NN] → restore_verification.id
check_code : TEXT [NN]
status : TEXT [NN]
checked_at : TEXT [NULL]
expected_value : TEXT [NN]
actual_value : TEXT [NULL]
evidence_key : TEXT [NULL]
failure_note : TEXT [NULL]

恢复验证 restore_verification
id : INTEGER [PK, NN]
backup_id : INTEGER [FK, NN] → backup_run.id
copy_id : INTEGER [FK, NN] → backup_copy.id
verified_by : INTEGER [FK, NN] → user_account.id (G13)
trigger_kind : TEXT [NN]
status : TEXT [NN]
isolated_path : TEXT [NN]
started_at : TEXT [NULL]
finished_at : TEXT [NULL]
source_snapshot_at : TEXT [NN]
recovery_target_at : TEXT [NN]
application_version : TEXT [NN]
result_note : TEXT [NULL]

备份文件清单 backup_artifact
id : INTEGER [PK, NN]
backup_id : INTEGER [FK, NN] → backup_run.id
artifact_kind : TEXT [NN]
document_version_id : INTEGER [FK, NULL] → document_version.id (G04)
relative_path : TEXT [NN]
size_bytes : INTEGER [NN]
sha256 : TEXT [NN]
verification_status : TEXT [NN]
verified_at : TEXT [NULL]
result_note : TEXT [NULL]

备份副本 backup_copy
id : INTEGER [PK, NN]
backup_id : INTEGER [FK, NN] → backup_run.id
location_kind : TEXT [NN]
storage_key : TEXT [NN]
manifest_sha256 : TEXT [NN]
status : TEXT [NN]
started_at : TEXT [NN]
verified_at : TEXT [NULL]
removed_at : TEXT [NULL]
removal_reason : TEXT [NULL]
verified_by : INTEGER [FK, NULL] → user_account.id (G13)
error_code : TEXT [NULL]

备份任务 backup_run
id : INTEGER [PK, NN]
run_no : TEXT [UQ, NN]
trigger_kind : TEXT [NN]
requested_by : INTEGER [FK, NULL] → user_account.id (G13)
responsible_staff_id : INTEGER [FK, NN] → staff.id (G01)
status : TEXT [NN]
started_at : TEXT [NULL]
snapshot_at : TEXT [NULL]
finished_at : TEXT [NULL]
schema_version : INTEGER [NULL]
application_version : TEXT [NULL]
manifest_key : TEXT [NULL]
manifest_sha256 : TEXT [NULL]
error_code : TEXT [NULL]
created_at : TEXT [NN]